

TECHNICAL REPORT

Credit Card Fraud Detection Using Logistic Regression and Decision Tree

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Data source: [Credit Card Dataset](#)

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Abstract

This project used R to analyze credit card transactions and compare Logistic Regression and Decision Tree models for fraud detection. The analysis included data exploration, feature scaling, visualization, model training, confusion matrices, ROC curves, AUC values, and DeLong's paired test. Logistic Regression achieved an AUC of 0.8257 and 99.38% accuracy, while the Decision Tree achieved an AUC of 0.7190 and 99.47% accuracy. Although the Decision Tree had slightly higher accuracy, Logistic Regression performed better at distinguishing fraudulent from legitimate transactions. The very small p-value from DeLong's test confirmed that the difference between the two models was statistically significant.

Summary

This project examined whether transaction amount can help distinguish fraudulent credit card transactions from normal transactions. The workflow included data exploration, feature standardization, a 75/25 training-test split, Logistic Regression, a Decision Tree classifier, confusion matrices, ROC curves, AUC values, repeated validation, and a paired DeLong test.

The final model comparison showed that Logistic Regression achieved an AUC of 0.8257, compared with 0.7189 for the Decision Tree. Their reported accuracies were 0.9938 and 0.9947, respectively. Although the Decision Tree produced slightly higher overall accuracy, Logistic Regression demonstrated stronger discrimination. The paired DeLong test returned $p < 2.2e-16$, indicating that the difference between the ROC curves was statistically significant.

1. Introduction

Fraud detection is a binary classification problem in which the objective is to separate fraudulent transactions from legitimate transactions. Because fraud cases are rare relative to normal transactions, accuracy alone can be misleading. A model can classify nearly every case as normal and still appear highly accurate. For this reason, the analysis emphasized ROC curves and AUC in addition to confusion matrices and accuracy.

The principal predictor used in the submitted models was transaction amount. The response variable was `is_fraud`, coded as 0 for a normal transaction and 1 for fraud.

2. Data Exploration

The training and testing files were imported and inspected for dimensions, feature names, missing values, summary statistics, variance, standard deviation, and transactions above the mean amount. The uploaded output reported no missing values. The amount variable was strongly right-skewed, with many low-value transactions and a small number of very large transactions.

The training violin plot shows that the two classes have different amount distributions. Fraudulent transactions are concentrated in several amount ranges, while normal transactions are more heavily concentrated at lower values and include a long upper tail. The log-scaled vertical axis makes these differences easier to see.

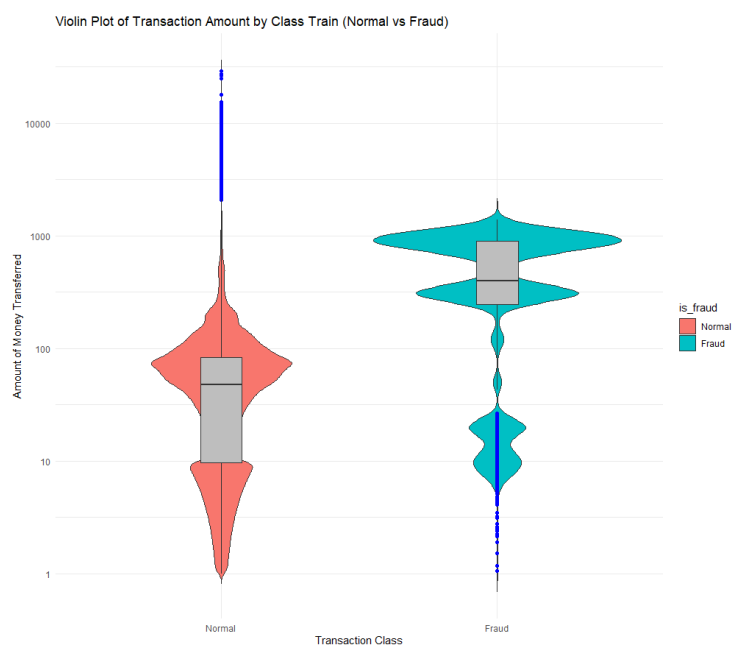


Figure 1. Distribution of transaction amounts by class in the training dataset.

The testing violin plot displays the same general pattern. The distributions are not normal; they are asymmetric and contain outliers. This supports the decision to standardize the amount feature and to use classification methods that do not require the predictor itself to be normally distributed.

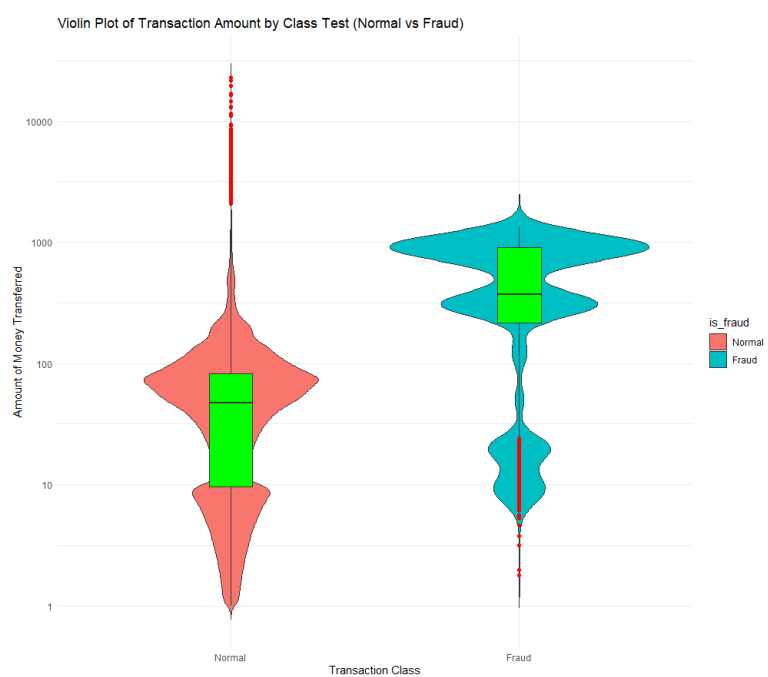


Figure 2. Distribution of transaction amounts by class in the testing dataset.

What's the shape of the violin plot? Is it a normal distribution?

The violin plot does not show a normal distribution. Instead, the transaction amounts are right-skewed, especially for the Normal class. Most transactions are concentrated at lower amounts, while a smaller number of transactions have very high amounts. This creates a long tail and several outliers, which makes the data not evenly distributed around the mean.

3. Data Preprocessing

The amount feature was standardized using the R `scale()` function. Standardization centers the variable around zero and expresses observations in standard-deviation units. A 75% training and 25% testing split was then created using a fixed seed for reproducibility.

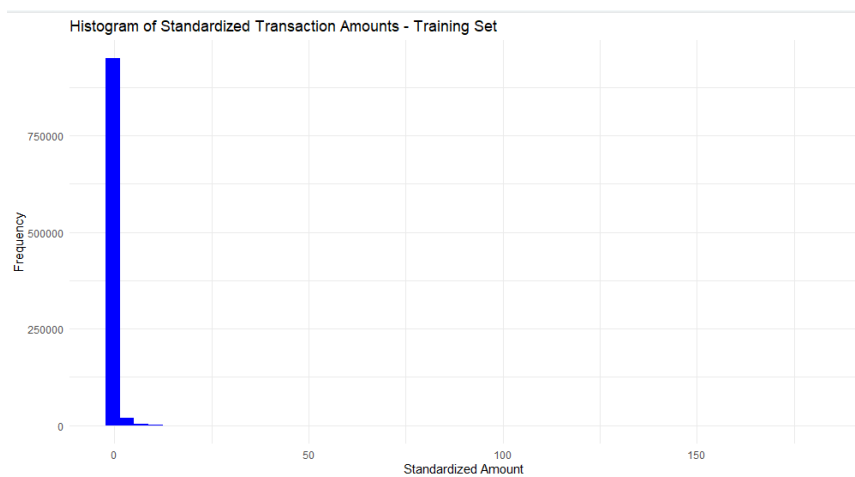


Figure 3. Histogram of standardized transaction amounts in the training set.

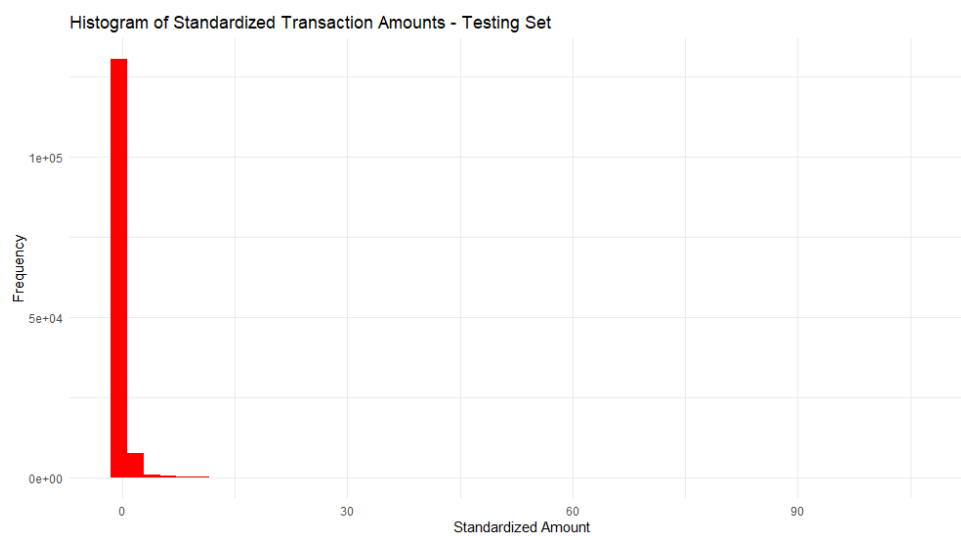


Figure 4. Histogram of standardized transaction amounts in the testing set.

Both histograms remain strongly right-skewed after standardization. Scaling changes the units and location of the feature, but it does not make a skewed variable normally distributed. The large concentration near the lower end reflects the fact that most transactions are relatively small.

4. Relationship Between Amount and Fraud Class

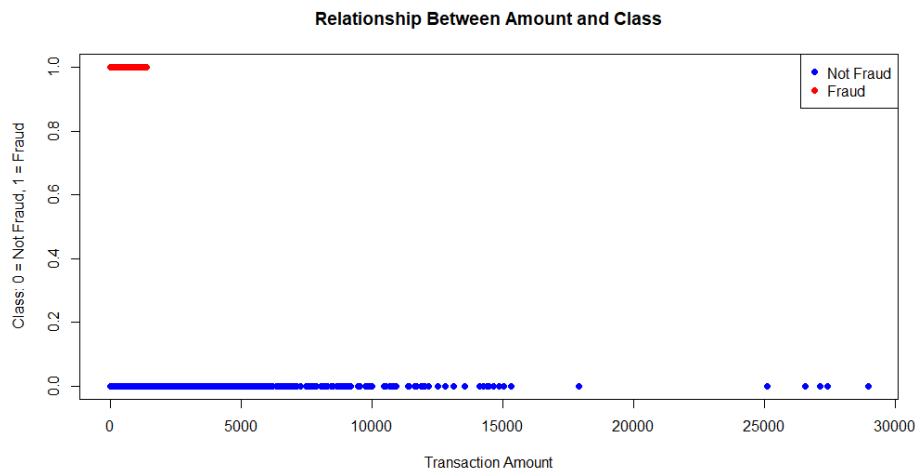


Figure 5. Scatter plot of transaction amount against the fraud class.

```
> train_set <- ccd[sample_index, ]  
> test_set <- ccdt[sample_index, ]  
> 
```

5. Logistic Regression Model

A binary Logistic Regression model was fitted using standardized transaction amount as the predictor. The model estimated the probability that each test observation belonged to the fraud class. A threshold of 0.50 was used to convert probabilities into predicted classes.

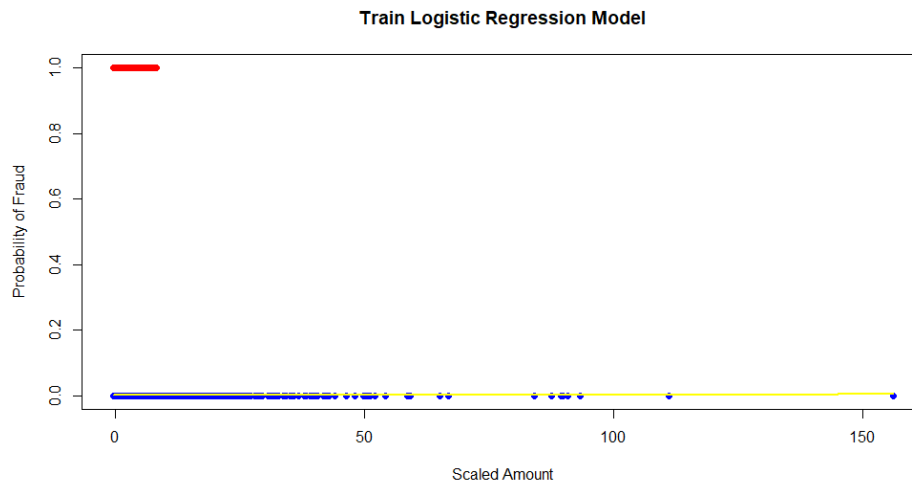


Figure 6. Logistic Regression fit on the training data.

```
> plot(
+   train_set$amt_scaled,
+   train_set$is_fraud,
+   main = "Train Logistic Regression Model",
+   xlab = "Scaled Amount",
+   ylab = "Probability of Fraud",
+   pch = 19,
+   col = ifelse(train_set$is_fraud == 1, "red", "blue")
+ )
> curve(
+   predict(
+     logistic_model,
+     newdata = data.frame(amt = x),
+     type = "response"
+   ),
+   add = TRUE,
+   col = "yellow",
+   lwd = 2
+ )
> |
```

The plots illustrate the fitted fraud probability across transaction amounts. Because fraud is rare and the model uses only one predictor, most predicted probabilities remain low. This explains why a 0.50 cutoff can produce few or no positive fraud predictions even when the ranking performance measured by AUC is acceptable.

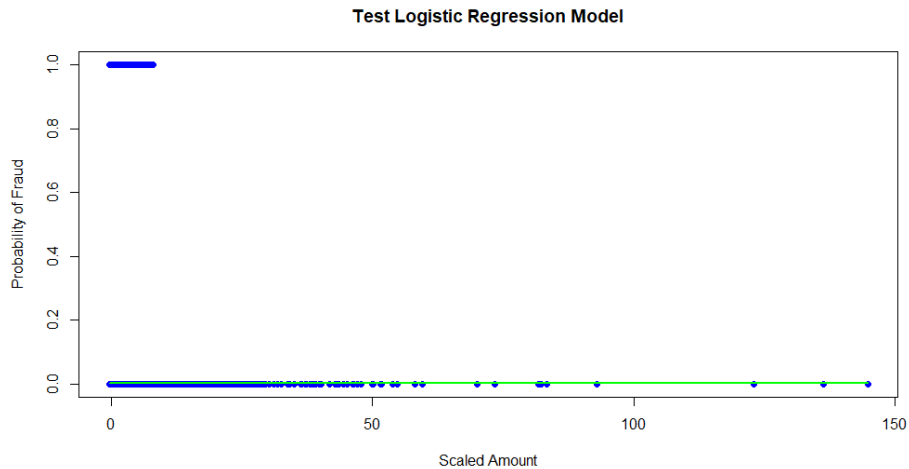


Figure 7. Logistic Regression fit on the testing data.

```
+ )
+ > plot(
+   test_set$amt_scaled,
+   test_set$is_fraud,
+   main = "Test Logistic Regression Model",
+   xlab = "Scaled Amount",
+   ylab = "Probability of Fraud",
+   pch = 19,
+   col = ifelse(train_set$is_fraud == 1, "red", "blue")
+ )
+ > curve(
+   predict(
+     logistic_model,
+     newdata = data.frame(amt = x),
+     type = "response"
+   ),
+   add = TRUE,
+   col = "green",
+   lwd = 2
+ )
+ > }
```

The standalone Logistic Regression analysis reported an AUC of 0.8379. Across ten different train-test splits, the median AUC was 0.8324515. These results suggest that the model has reasonably stable ranking ability, although it is not sufficient for high-sensitivity fraud detection when used with transaction amount alone.

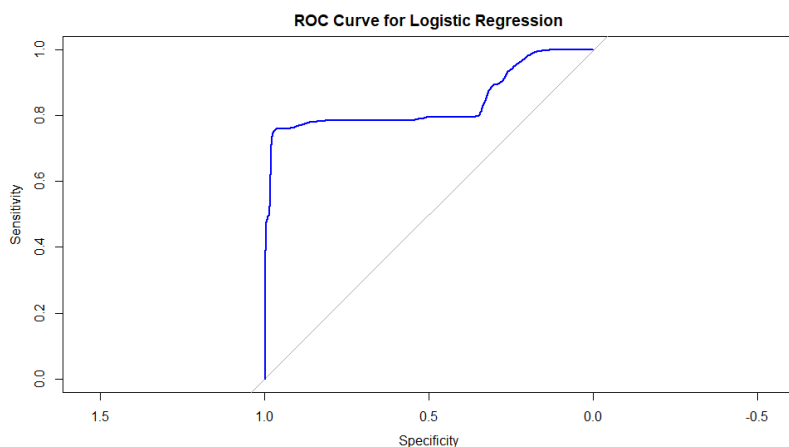


Figure 8. ROC curve for the standalone Logistic Regression run.


```

> roc_logistic <- roc(
+   response = test_set$is_fraud,
+   predictor = test_set$predicted_probability
+ )
Setting levels: control = 0, case = 1
Setting direction: controls < cases

> plot(
+   roc_logistic,
+   main = "ROC Curve for Logistic Regression",
+   col = "blue",
+   lwd = 2
+ )
>

```

Show AUC results table

auc_results

```

> # Show AUC results table
> auc_results
  Run Seed      AUC
1    1  101 0.8344520
2    2  202 0.8293065
3    3  303 0.8334391
4    4  404 0.8296150
5    5  505 0.8312194
6    6  606 0.8320563
7    7  707 0.8341833
8    8  808 0.8288917
9    9  909 0.8357003
10   10 1001 0.8328468

```

Figure 9. AUC values from ten Logistic Regression runs and the median AUC.

Calculate median AUC

```
median_auc <- median(auc_results$AUC)
```

median_auc

```

> # Calculate median AUC
> median_auc <- median(auc_results$AUC)
> median_auc
[1] 0.8324515
>

```

6. Decision Tree Model

A Decision Tree classifier was trained on the same standardized amount feature. The tree divided transactions using a sequence of amount thresholds. Each terminal node assigned a predicted class based on the transactions reaching that node.

The tree structure demonstrates a non-linear decision process. Several splits occur at relatively high standardized amounts, showing that the algorithm searched for local ranges in which fraud prevalence differed. However, a single numerical feature limits the tree's ability to represent broader behavioural patterns associated with fraud.

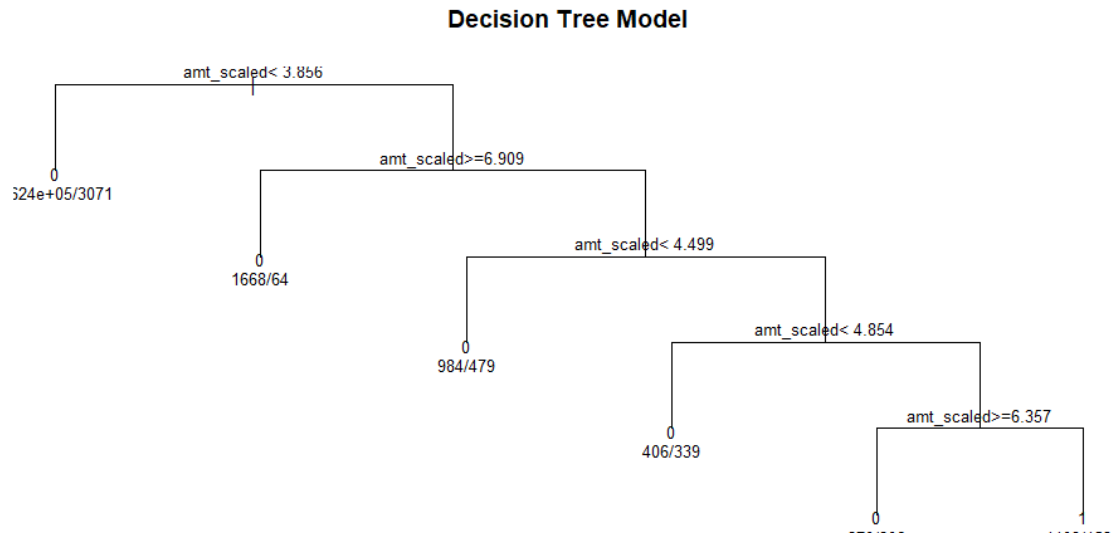


Figure 10. Decision Tree generated from standardized transaction amount.

```

> # Plot decision tree using base R
> plot(decision_tree_model, uniform = TRUE, main = "Decision Tree Model")
> text(decision_tree_model, use.n = TRUE, cex = 0.8)
> |

```

Decision tree confusion matrix and accuracy

```

> decision_tree_confusion_matrix <- table(
+   Actual = test_set$is_fraud,
+   Predicted = test_set$dt_predicted_class
+ )

> decision_tree_confusion_matrix
      Predicted
Actual      0      1
0    321973    388
1     1337    471

> |

> decision_tree_accuracy <- sum(diag(decision_tree_confusion_matrix)) / sum(decision_tree_confusion_matrix)
> decision_tree_accuracy
[1] 0.9946787
> |

```

7. Model Performance Comparison

In the final paired comparison, Logistic Regression achieved an AUC of 0.8257013, while the Decision Tree achieved an AUC of 0.7189060. The Logistic Regression ROC curve lies above the Decision Tree curve over much of the operating range, indicating better discrimination between fraud and normal transactions.

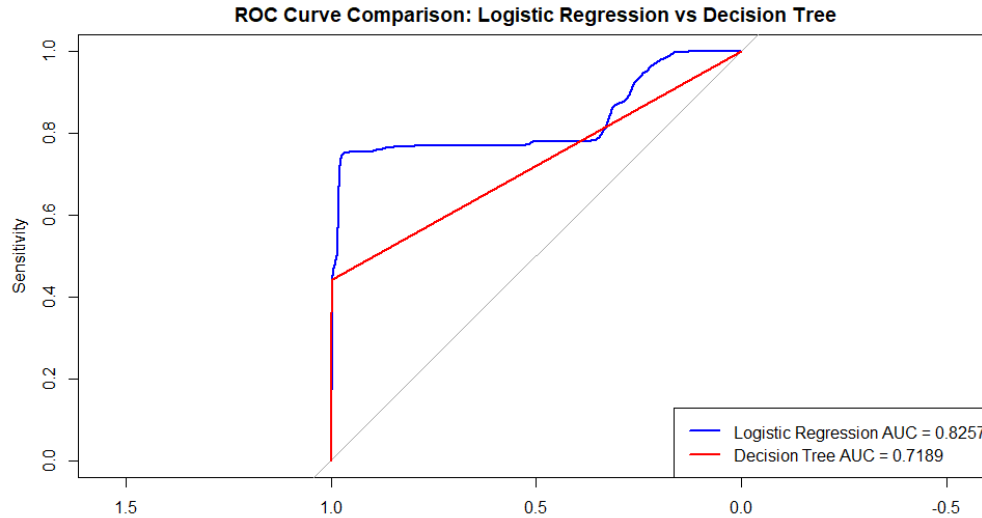


Figure 11. ROC comparison of Logistic Regression and Decision Tree.

```
> plot(
+   roc_logistic,
+   col = "blue",
+   lwd = 2,
+   main = "ROC Curve Comparison: Logistic Regression vs Decision Tree"
+ )
> lines(
+   roc_decision_tree,
+   col = "red",
+   lwd = 2
+ )
> legend(
+   "bottomright",
+   legend = c(
+     paste("Logistic Regression AUC =", round(as.numeric(auc_logistic), 4)),
+     paste("Decision Tree AUC =", round(as.numeric(auc_decision_tree), 4))
+   ),
+   col = c("blue", "red"),
+   lwd = 2
+ )
>
```

7.1 Summary of Final Model Results

The Decision Tree had slightly higher accuracy, but its AUC was substantially lower. This apparent contradiction is caused by class imbalance: predicting the majority normal class correctly can produce very high accuracy without reliably identifying fraud. AUC is therefore the more informative comparison metric in this setting.

```
> auc_comparison <- data.frame(
+   Model = c("Logistic Regression", "Decision Tree"),
+   AUC = c(as.numeric(auc_logistic), as.numeric(auc_decision_tree)),
+   Accuracy = c(logistic_accuracy, decision_tree_accuracy)
+ )
> auc_comparison
  Model      AUC Accuracy
1 Logistic Regression 0.8257013 0.9937872
2 Decision Tree 0.7189060 0.9946787
>
```

8. Hypothesis Testing

A paired DeLong test was used because both ROC curves were calculated from predictions on the same test observations. The hypotheses were: H_0 - the two population AUC values are equal; H_1 - the AUC values are different.

```
> roc_test_result <- roc.test(
+   roc_logistic,
+   roc_decision_tree,
+   method = "deLong",
+   paired = TRUE
+ )
> roc_test_result

      DeLong's test for two correlated ROC curves

data:  roc_logistic and roc_decision_tree
Z = 16.509, p-value < 2.2e-16
alternative hypothesis: true difference in AUC is not equal to 0
95 percent confidence interval:
 0.09411633 0.11947429
sample estimates:
AUC of roc1 AUC of roc2
 0.8257013  0.7189060
> |
```

Figure 12. Paired DeLong test output for the two correlated ROC curves.

The test produced $Z = 16.509$ and $p < 2.2e-16$. At the 0.05 significance level, the null hypothesis is rejected. The evidence therefore indicates a statistically significant difference between the two AUC values. Based on the observed AUCs, Logistic Regression performed significantly better than the Decision Tree in ranking fraudulent transactions above normal transactions.

9. Limitations

- Only transaction amount was used as a predictor, although the dataset contains many additional potentially informative features.
- The default 0.50 classification threshold is not optimized for a rare-event problem and produced weak fraud recall.
- Accuracy is inflated by class imbalance and should not be interpreted as evidence of strong fraud detection.
- The milestone outputs were generated from different intermediate splits, so the standalone Logistic Regression AUC differs slightly from the final paired-comparison AUC.
- No class weighting, resampling, precision-recall analysis, or cost-sensitive threshold selection was applied.

10. Recommendations

- Add behavioural and contextual predictors such as merchant category, time, location, and customer transaction history.
- Evaluate sensitivity, specificity, precision, recall, F1-score, and the precision-recall curve in addition to ROC/AUC.
- Tune the probability threshold according to the financial cost of missed fraud and false alerts.

- Use class weights or imbalance-handling methods and compare additional algorithms such as Random Forest and gradient boosting.
- Use cross-validation or a fixed holdout set to ensure that every model is compared under identical conditions.

Conclusion

The project successfully implemented an end-to-end fraud classification workflow in R. The analysis included exploration, scaling, visualization, model fitting, repeated evaluation, ROC/AUC analysis, and formal hypothesis testing. Logistic Regression produced stronger discriminatory performance than the Decision Tree, with final AUC values of 0.8257 and 0.7189. The paired DeLong test confirmed that this difference was statistically significant. However, the results also show that transaction amount alone is insufficient for reliable operational fraud detection, and future models should incorporate additional predictors and imbalance-aware evaluation methods.

References

An introduction to ROC analysis / pattern recognition letters. Available at:

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Hosmer Jr., D.W., Lemeshow, S. and Sturdivant, R.X. (2013) applied logistic regression. 3rd edition, John Wiley & Sons, Hoboken, NJ. - references - scientific research publishing. Available at:

<https://www.scirp.org/reference/referencespapers?referenceid=2061497>.

Intelligent financial fraud detection: A comprehensive review - charles sturt university research output. Available at: <https://researchoutput.csu.edu.au/en/publications/intelligent-financial-fraud-detection-a-comprehensive-review>.